Documentation

Questions to be Answered

1. What defines a high quality wine
2. What chemical properties are essential for that
3. How can we act/alter them (assortment decisions)

Data

-Total 1599 entries

-12 columns - all of the features are float , 10 categories for the quality

-lots of class imbalance in teh data set (mainly 5 and 6 holds all the position)

- alcohol affects the quality most of all the other features , than comes sulphate and after that citric acid .

- There are many features which might affect the quality negatively

Feature Importance

	Feature	Importance
10	alcohol	0.272652
9	sulphates	0.151100
1	volatile acidity	0.108390
6	total sulfur dioxide	0.077636
4	chlorides	0.070043
8	pH	0.061484
3	residual sugar	0.058068
7	density	0.052488
0	fixed acidity	0.052336
2	citric acid	0.048978
5	free sulfur dioxide	0.046824

Analysis

Although we have the less import features for the quality but stil we need them in order to make the correct low quality features

while a feature might have an inverse relationship with the target (negative correlation), it can still be a highly *important* predictor, and its importance score will be a positive value reflecting that contribution.

Analysis

High-quality wines tend to have:
- Higher alcohol content
- Lower volatile acidity
- Balanced sulphates

Lower-quality wines often show:
- High acidity
- Higher density

Analysis

1. Key Drivers of Wine Quality

- Alcohol ↑ → Higher rating
- Volatile acidity ↑ → Lower rating

2. Model Insight

- Model explains ~X% variance
- Reliable for trend detection, not perfect prediction

3. Business Recommendation

- Prioritize wines with:
  - higher alcohol
  - lower acidity
- Focus assortment on mid-to-high quality segment

Improving the model

Balanced the classes

Made the search CV for better parameters

Gradient boosting outperforms the randomizedSearchCv so trying the xgboost

While the performance on the training data after hypertuning the parameters was quiet good (0.87) but on the testing it hasn´t done any significantly better than only Smote data . so the overall accuracy we got is still 0.67

Recommendation

1.Red wines with alcohol > 11%, sulphates > 0.65 g/L, and volatile acidity < 0.4 g/L have a 73% probability of being rated high quality. I recommend setting these as minimum thresholds for assortment selection.

Best Recommendation according to data

	High Quality (7-8)	Low Quality (3-5)	Difference
fixed acidity	8.847005	8.142204	0.704800
volatile acidity	0.405530	0.589503	-0.183973
citric acid	0.376498	0.237755	0.138742
residual sugar	2.708756	2.542070	0.166686
chlorides	0.075912	0.092989	-0.017077
free sulfur dioxide	13.981567	16.567204	-2.585637
total sulfur dioxide	34.889401	54.645161	-19.755760
density	0.996030	0.997068	-0.001038
pH	3.288802	3.311653	-0.022851
sulphates	0.743456	0.618535	0.124921
alcohol	11.518049	9.926478	1.591571

Visualization